

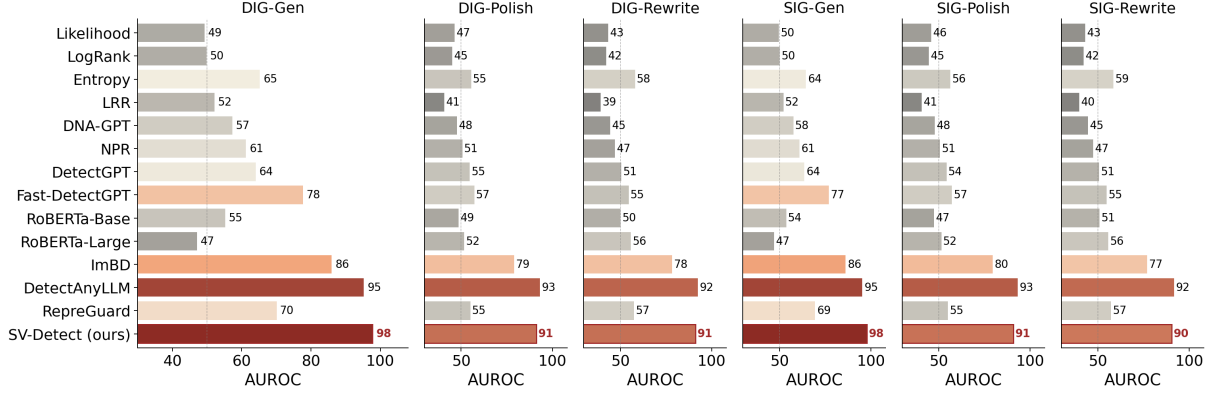


Figure 4: In-distribution performance on MIRAGE across the GENERATE, POLISH, and REWRITE tasks under the DIG (left) and SIG (right) settings.

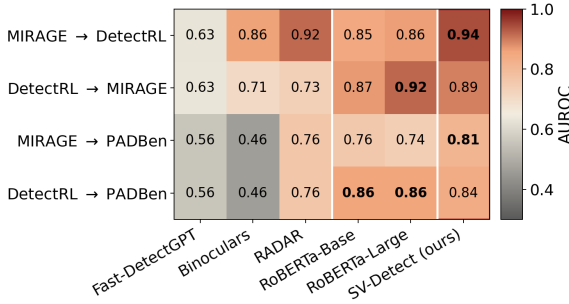


Figure 5: Cross-benchmark transfer. Grand-mean AUROC for each transfer direction (rows) and detector (columns)

uate it on a different target benchmark, without retraining, threshold calibration, or access to target-benchmark data.

We evaluate four transfer directions, using MIRAGE and DetectRL as source benchmarks and DetectRL, MIRAGE, and PADBen as targets: (i) MIRAGE → DetectRL, (ii) DetectRL → MIRAGE, (iii) MIRAGE → PADBen, and (iv) DetectRL → PADBen. In each case, the detector is trained only on the source benchmark’s training pool described in Sec. 4.1: for MIRAGE, we use the default 500 POLISH training pairs, and for DetectRL, we fit a single steering direction per layer using the union of the four DetectRL Multi-Domain training subsets. We evaluate on the target benchmark’s test split without retraining, threshold calibration, or access to any target-benchmark data.

We compare against the strongest zero-shot detectors reported across the DetectRL and PADBen baselines: Fast-DetectGPT (Bao et al., 2023), Binoculars (Hans et al., 2024), and RADAR (Hu et al., 2023). We also include RoBERTa-Base and RoBERTa-Large models fine-tuned on the

same source pool as SV-Detect, giving a matched-supervision comparison with strong discriminative classifiers. We report macro-averaged AUROC across the constituent settings of each target benchmark: 13 configurations for DetectRL (4 Multi-Domain subsets, 4 Multi-LLM generators, and 5 Multi-Attack categories), 6 sentence-pair tasks for MIRAGE, and 5 sentence-pair regimes for PADBen. Full per-cell breakdowns are provided in Appendix Tables 12, 13, and 14.

Figure 5 summarizes the results. Overall, SV-Detect exhibits stable and consistently strong transfer across benchmarks: it is the best or second-best method in every transfer direction. The strongest results are obtained when MIRAGE is used as the source benchmark. SV-Detect is the top-performing method on both MIRAGE → DetectRL and MIRAGE → PADBen. When trained on DetectRL, SV-Detect also transfers well, substantially outperforming all zero-shot detectors on both MIRAGE and PADBen. However, RoBERTa-Large trained on the same DetectRL pool performs slightly better in aggregate on these two targets.

4.3 Ablation studies

Ablation summary. Fig. 6a and Fig. 6b summarize three design choices in SV-Detect on DetectRL Multi-Domain transfer: choice of frozen LM backbone, the downstream classifier and the steering vector construction method. For each choice, we report three complementary quantities: mean in-distribution AUROC, mean transfer AUROC (averaged over off-diagonal pairs), and worst-case transfer AUROC (minimum over transfer pairs). This compact view captures not only average performance, but also robustness under the hardest distribution shifts.

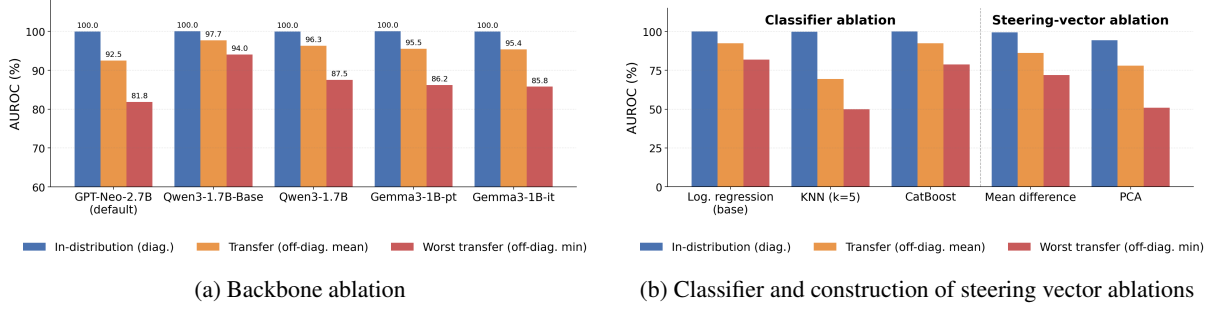


Figure 6: Ablation summary on DetectRL Multi-Domain transfer.

Choice of downstream classifier. Among the classifiers we consider, logistic regression is the most reliable overall and is therefore used as the default detector throughout the paper. While CatBoost remains competitive in-distribution, it is consistently weaker under transfer, especially in the worst case. KNN is substantially less stable and often collapses to near-chance performance on off-diagonal evaluations. This suggests that the main strength of SV-Detect lies in the steering-based representation itself rather than in a highly expressive nonlinear classification head.

Choice of steering vector construction. The choice of steering vector construction has an even stronger effect. Logistic-regression-based steering vectors are clearly the most robust, achieving both the strongest average transfer and the strongest worst-case transfer. In contrast, mean-difference and PCA-based directions degrade much more severely, with worst-case transfer often approaching chance level. This shows that explicitly learning a discriminative direction in activation space is substantially more effective than relying on unsupervised variance directions or raw class-mean differences, and justifies our use of LogReg-based steering in the main experiments.

Choice of frozen LM backbone. We use GPT-Neo as the default backbone in the main experiments for fair comparison with prior work. On Fig. 6a, we compare several alternative backbones of similar scale. The results show that SV-Detect is not tied to GPT-Neo: all tested backbones perform strongly and, in fact, better than GPT-Neo at the worst-case transfer, with Qwen3-1.7B-Base being the most robust under transfer. Thus, backbone choice affects robustness, but the overall steering vector approach generalizes beyond the particular LM used in the main paper. More detailed analysis is provided in Appendix Sec. C.

5 What do the steering vectors detect?

We study both *where* SV-Detect extracts signal from and *what* lexical or stylistic patterns are associated with the learned directions.

Our detector is a logistic regression on layer-wise projection scores, $\mathbf{s}(x) = (s_1(x), \dots, s_L(x))$, with one feature per layer on DetectRL and one feature per layer-direction pair on MIRAGE. To identify the most important layers, we inspect the magnitude-weighted coefficients $|w_l| \cdot \sigma_l$. On MIRAGE, the signal concentrates in the final layers (L29–31), and on DetectRL it is more distributed, with peaks at L0–2, L14–18, and L21. A more detailed layer-wise analysis is given in supplementary Section D.

Logit-lens interpretation. For the top-contributing layers, we interpret the steering vector v_l by projecting it through the final layer norm and LM head:

$$\text{logits}_l = \text{LM_head}(\text{LN}_f(v_l)).$$

Top-ranked tokens correspond to the $+v_l$ direction (AI-generated text side), and bottom-ranked tokens to $-v_l$ (human-text side). This is a standard logit-lens probe applied to a learned direction.

Lexical signatures. Figure 8a shows that the learned directions are lexically interpretable, though differently on MIRAGE and DetectRL. On MIRAGE, the $+v_l$ direction at L29–31 surfaces content-bearing, often technical or formal fragments, while the $-v_l$ direction is dominated by punctuation-heavy fragments together with a few proper nouns and specific lexical items. On DetectRL, the $+v_l$ direction at L14–18 reflects a more polished LLM-like register, while the $-v_l$ direction surfaces more colloquial and topical fragments. Thus, the steering vectors align with meaningful lexical and stylistic differences between human-written and machine-generated text.

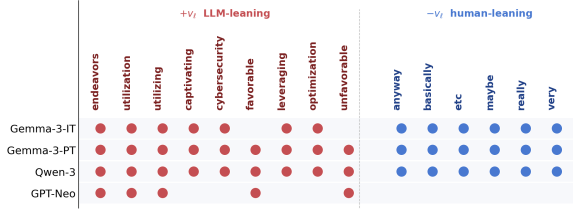
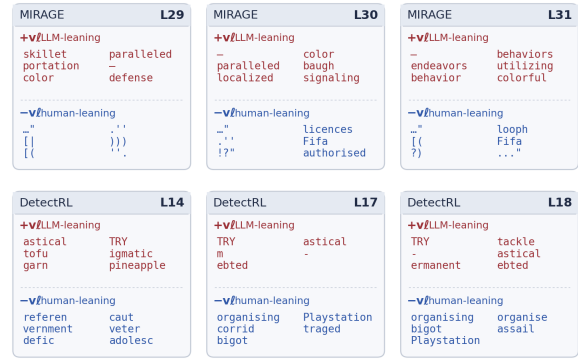


Figure 7: Consensus tokens across the four LMs. A colored dot indicates that the token appears in that LM’s pooled top-token set.

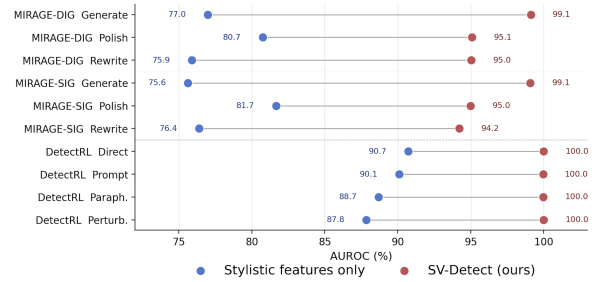
Beyond surface cues. To quantify how much of the signal is explained by simple stylistic features, we train a logistic regression on interpretable regex-based counts derived from the logit-lens analysis. The full setup is given in the Appendix Sec. D. These features achieve 76-82 AUROC on MIRAGE and 88-91 AUROC on DetectRL, but the full steering vector pipeline improves over them by 13-24 AUROC points across settings (Figure 8b). This suggests that the learned directions capture not only recognizable surface markers, but also additional signal in the hidden representations.

Cross-LM logit-lens analysis. We further test whether the lexical signal identified above is specific to GPT-Neo or shared across reference LMs. We retrain the steering vector pipeline on four LMs (GPT-Neo-2.7B, Qwen3-1.7B (Yang et al., 2025), Gemma-3-1B-pt, and Gemma-3-1B-it (Team et al., 2025)) and decode the resulting dense steering vectors as in Sec. D.2. Figure 7 summarizes the result. Despite tokenizer- and corpus-specific differences, the same broad positive-side register recurs across models: all four surface endeavors, utilization, and utilizing, while the negative side is consistently more casual and discourse-like. This supports the view that SV-Detect captures a broader LLM-associated register rather than a GPT-Neo-specific artifact.

Per-token contribution. SV-Detect also admits a token-level decomposition of the final score. For a fixed layer l , we project each token activation at that layer onto the corresponding steering vector and use the resulting signed value as that token’s contribution to the detector output. Figure 1 visualizes this signal. LLM-generated text tends to exhibit localized spans with consistently strong positive evidence, whereas human-written text shows weaker and more heterogeneous contributions. This qualitative contrast is consistent with the analyses above.



(a) Top tokens at the most contributing layers.



(b) In-domain AUROC of a regex detector vs. the full pipeline.

Figure 8: Steering vector interpretation.

6 Conclusion

We introduced **SV-Detect**, an AI-generated text detector based on steering vectors extracted from the hidden representations of a frozen language model. By representing each text through its alignment with layer-wise human-vs-AI directions and training lightweight classifier on top, SV-Detect provides a simple, interpretable alternative to text-level score-based and fully supervised detectors.

Experiments show that SV-Detect performs strongly both in-distribution and under challenging transfer settings, including shifts across domains, source models, attack families, and machine-editing transformations. It also transfers effectively across benchmarks, suggesting that it captures a benchmark-independent signal of AI-generated text.

Our interpretation analyses further show that the learned directions align with meaningful lexical and stylistic cues while also encoding substantial additional representation-level signal. Overall, these results support a representation-space view of AI-generated text detection: machine-generated text can be identified not only from its surface form, but also from stable directions in hidden-state space.